

CSE470: Software Engineering

SEC - 9

Semester: **Fall 2025**
Examination: **Quiz 3**

Time: **20 minutes**
Full marks: **15**

Name: _____

ID: _____

Section: **9**

Imagine you are a software developer for your local sports news outlet. Now you have been tasked to develop their media system.

They only cover two home football teams: Manchester United and Manchester City. They have a very large number of followers. The followers can individually choose to get updates about ManU or ManCity. If a fan chooses to get updates from ManU, that fan should not get updates about Man City. One fan can choose to follow both teams for updates, in which case that fan will receive updates about both teams. The fans can unfollow each of the teams if they want as well. When a fan unfollows any team, that fan should not receive any update about that team. Aside from this, your boss is concerned about possible data copyright issues therefore you need to make sure that there are no data inconsistency issues from your news outlet to your fans. To complete this task your teams software architect has given you the following directives:

- There will be 2 classes: NewsOutlet, Fan
- Key Methods for NewsOutlet Class: addFan, teamUpdate, notifyfan, removeFan
- Key Methods for Fan Class: followTeam, receive, unfollowTeam

PS: You may need to use at least 1 additional method to better suit the client requirements.

1. What design pattern/s should be used here? [1 Mark]

Observer + Singleton

2. Implement the required classes and methods. [14 Marks]

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BRAC University
Department of Computer Science and Engineering (CSE)

CSE470: Software Engineering

SEC - 19

Semester: **Fall 2025**
Examination: **Quiz 2**

Time: **20 minutes**
Full marks: **15**

Name: _____ ID: _____ Section: **19**

Imagine you are a software developer for your local Study Abroad agency. Now you have been tasked to develop their updated application system.

The Local Agency's current software initially takes the name, address and the dob of the applicant as a part of registration. After the registration it asks the applicant for its college information, school information as well as the ielts score separately. Currently your local agency plans to support Australian University admissions. Australian universities like other countries have their own set of application instructions and formats. Which is why your agency has asked you to develop a new software that will follow those guidelines using data from the existing software. Australian universities take the name, address and dob as initial registration information. It takes both college and university information in the same function. It also accepts English proficiency test scores. You also need to make sure the data is consistent during the process. To complete this task your teams software architect has given you the following directives:

There will be 3 Classes: AustraliaEdu, LocalEdu and DataMigration

Key Methods for AustraliaEdu: personalDetails, setEducationalDetails and englishTest

Key Methods for LocalEdu: naamPorichoy, setUniversity, setCollege, ielts

PS1: You need to design the DataMigration class yourself.

PS2: You may need to use at least 1 additional method to better suit the client requirements.

1. What design pattern/s should be used here? [1 Mark]

Adapter Pattern + Singleton

2. Implement the required classes and methods. [14 Marks]

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